A Valuation Framework for Sealed Products in Collectible Card Markets:

Expected Value, No-Arbitrage Pricing, and the Sealed–Opened Spread

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Abstract

We develop a probabilistic valuation model for sealed collectible card game products—specifically Magic: The Gathering—comparing the market price of a sealed product with the expected value of its contents if opened and sold individually on the secondary market. We define the *sealed–opened spread* as the difference between these two quantities and derive a no-arbitrage condition under perfect market assumptions. We then discuss the role of transaction costs, variance, and liquidity in explaining persistent deviations from the theoretical benchmark. A generic numerical example illustrates the framework.

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1 Introduction

Collectible card games (CCGs) present an unusual asset pricing problem. A sealed product—a booster pack, a deck box, a collector's edition—contains a random assortment of cards drawn from a known collection according to a known (or approximately known) probability distribution. Each individual card has a market price determined by the secondary market. The sealed product itself also has a market price.

This creates a natural question: what is the relationship between the price of the sealed product and the expected value of its contents? Under what conditions should these coincide? And when they diverge, what explains the gap?

This paper formalizes these questions. We construct a model in which a sealed product is treated as a lottery over portfolios of individual cards, compute the expected value of opening the product, and compare it to the observed sealed price. We call the difference the *sealed–opened spread* and analyze the conditions under which it should be zero (no-arbitrage) and the market frictions that cause it to deviate.

The model applies to any sealed CCG product but is developed with Magic: The Gathering (MTG) in mind. MTG products vary in structure—draft boosters contain roughly 15 cards, commander preconstructed decks contain 100 cards, collector boosters contain 15 cards of higher average rarity—and we write the model generally enough to accommodate all of these.

2 Setup and Notation

Definition 2.1 (Collection)**.** *A* collection *(or* set*)* $\mathcal{S}$ *is a finite set of distinct cards,*

$$\mathcal{S} = \{c_1, c_2, \ldots, c_Z\},$$

where $Z = |\mathcal{S}|$ *is the total number of distinct cards in the collection.*

In MTG, each expansion set defines a collection. For example, a typical Standard-legal set might contain $Z \approx 250$–300 distinct cards, while a supplemental set designed for the secondary market might contain $Z \approx 300$–450 cards including bonus sheet reprints and special treatments.

Definition 2.2 (Rarity Partition)**.** *The collection* $\mathcal{S}$ *is partitioned by rarity into disjoint subsets,*

$$\mathcal{S} = \mathcal{C} \cup \mathcal{U} \cup \mathcal{R} \cup \mathcal{M},$$

where $\mathcal{C}$*,* $\mathcal{U}$*,* $\mathcal{R}$*,* $\mathcal{M}$ *denote the sets of commons, uncommons, rares, and mythic rares, respectively. We write* $C = |\mathcal{C}|$*,* $U = |\mathcal{U}|$*,* $R = |\mathcal{R}|$*,* $M = |\mathcal{M}|$*, so that* $Z = C + U + R + M$*.*

Definition 2.3 (Card Price)**.** *For each card $c_i \in \mathcal{S}$ and time $t \geq 0$, the* market price *$p_i(t) \in \mathbb{R}_{\geq 0}$ is the price at which card c_i can be sold on the secondary market at time t.*

In practice, $p_i(t)$ is the mid-price (average of the best bid and best ask) on a secondary marketplace such as TCGPlayer or CardMarket. The price is a function of time because demand, supply, tournament results, ban-list changes, and speculative activity cause prices to fluctuate.

Definition 2.4 (Sealed Product)**.** *A* sealed product *P associated with collection $\mathcal{S}$ is characterized by:*

(i) A number of card slots *$n \in \mathbb{N}$, representing the total number of cards contained in the product.*

(ii) A collection of slot-level probability distributions. For each slot $k \in \{1, \ldots, n\}$, a probability mass function

$$\boldsymbol{\pi}_k = (\pi_{k,1}, \pi_{k,2}, \ldots, \pi_{k,Z}), \quad \textit{where} \quad \pi_{k,i} = \Pr(\textit{card } c_i \textit{ occupies slot } k),$$

satisfying $\pi_{k,i} \geq 0$ for all i and $\sum_{i=1}^{Z} \pi_{k,i} = 1$.

The slot structure is important. In a typical MTG draft booster with $n = 15$ cards:

- 1 slot is the *rare/mythic rare slot*: it contains a rare card with probability $\approx 7/8$ and a mythic rare with probability $\approx 1/8$, each equally likely within its rarity class.
- 3 slots are *uncommon slots*: each draws uniformly from $\mathcal{U}$.
- 10 slots are *common slots*: each draws uniformly from $\mathcal{C}$.
- 1 slot is a *land/bonus slot* with its own distribution (foils, basic lands, etc.).

For a preconstructed commander deck, the situation is different: the contents are deterministic ($n = 100$ cards, fully specified), so $\pi_{k,i} \in \{0, 1\}$ for all k, i. In this case the "lottery" is degenerate.

Definition 2.5 (Sealed Price)**.** *The* sealed price *of product P at time t is denoted $V^{\mathrm{S}}(t) \in \mathbb{R}_{\geq 0}$.*

3 The Expected Value of Opening

3.1 Single-Slot Expected Value

For a single slot k at time t, the expected value of the card drawn is

$$\mathbb{E}[X_k(t)] = \sum_{i=1}^{Z} \pi_{k,i} \cdot p_i(t). \tag{1}$$

This is straightforward: the expected value of slot k is the probability-weighted sum of all possible card prices.

3.2 Product-Level Expected Value

Summing over all n slots, the *expected value of opening* the product is

$$V^{\mathrm{O}}(t) := \mathbb{E}\left[\sum_{k=1}^{n} X_k(t)\right] = \sum_{k=1}^{n}\sum_{i=1}^{Z} \pi_{k,i} \cdot p_i(t). \tag{2}$$

By linearity of expectation, this holds regardless of any dependencies across slots (e.g., collation rules that prevent duplicate rares within a single pack).

Remark 3.1. *If slots within a product are subject to a* no-duplicate constraint*—a common feature in MTG boosters where you typically cannot open two copies of the same rare in one pack—then the joint distribution across slots is not a product of the marginals. However, linearity of expectation still applies: the expected value of the sum equals the sum of the expected values, even under dependence. This is a key simplification.*

3.3 Rarity-Based Simplification

When the probability distribution within a slot is uniform over a rarity class (as is standard in MTG), the expected value simplifies considerably. Define the *average price by rarity* at time t:

$$\bar{p}_{\mathcal{C}}(t) = \frac{1}{C}\sum_{c_i \in \mathcal{C}} p_i(t), \tag{3}$$

$$\bar{p}_{\mathcal{U}}(t) = \frac{1}{U}\sum_{c_i \in \mathcal{U}} p_i(t), \tag{4}$$

$$\bar{p}_{\mathcal{R}}(t) = \frac{1}{R}\sum_{c_i \in \mathcal{R}} p_i(t), \tag{5}$$

$$\bar{p}_{\mathcal{M}}(t) = \frac{1}{M}\sum_{c_i \in \mathcal{M}} p_i(t). \tag{6}$$

Then for a standard draft booster, the expected value of the rare/mythic slot is

$$\mathbb{E}[X_{\text{rare slot}}(t)] = \frac{7}{8} \cdot \bar{p}_{\mathcal{R}}(t) + \frac{1}{8} \cdot \bar{p}_{\mathcal{M}}(t), \tag{7}$$

and the total product expected value becomes

$$V^{\mathrm{O}}(t) = \underbrace{\frac{7}{8}\,\bar{p}_{\mathcal{R}}(t) + \frac{1}{8}\,\bar{p}_{\mathcal{M}}(t)}_{\text{rare/mythic slot}} + \underbrace{3\,\bar{p}_{\mathcal{U}}(t)}_{\text{uncommon slots}} + \underbrace{10\,\bar{p}_{\mathcal{C}}(t)}_{\text{common slots}} + \underbrace{\mathbb{E}[X_{\text{bonus}}(t)]}_{\text{bonus slot}}. \tag{8}$$

In practice, common cards have near-zero secondary market value (often $p_i(t) <$ \$0.10), so the expected value of a booster is dominated by the rare/mythic slot plus any high-value uncommons.

4 The Sealed–Opened Spread

Definition 4.1 (Sealed–Opened Spread)**.** *The* sealed–opened spread *at time t is*

$$\Delta(t) \coloneqq V^{\mathrm{S}}(t) - V^{\mathrm{O}}(t). \tag{9}$$

When $\Delta(t) > 0$, the sealed product trades at a premium to its expected contents—a buyer of the sealed product is paying more than the expected value of the cards inside. When $\Delta(t) < 0$, the sealed product trades at a discount—an arbitrageur could in principle buy the product, open it, and sell the cards for an expected profit.

4.1 The No-Arbitrage Condition

Assumption 4.1 (Perfect Market)**.** *Assume:*

(a) No transaction costs: cards can be bought and sold at the same price $p_i(t)$.

(b) No liquidity constraints: any quantity of any card can be sold instantly.

(c) Risk neutrality: agents care only about expected value, not variance or higher moments.

(d) Symmetric information: all agents know the probability distributions $\{\boldsymbol{\pi}_k\}$ and prices $\{p_i(t)\}$.

(e) Frictionless entry: any agent can buy sealed product and sell cards, or vice versa.

Proposition 4.1 (No-Arbitrage Pricing)**.** *Under Assumption 4.1, the sealed–opened spread is zero:*

$$\Delta(t) = 0 \quad \Longleftrightarrow \quad V^{\mathrm{S}}(t) = V^{\mathrm{O}}(t).$$

Argument. Suppose $\Delta(t) < 0$, i.e., $V^{\mathrm{S}}(t) < V^{\mathrm{O}}(t)$. An agent could buy the sealed product at price $V^{\mathrm{S}}(t)$, open it, and sell the contents on the secondary market. The expected revenue is $V^{\mathrm{O}}(t) > V^{\mathrm{S}}(t)$, yielding an expected profit of $V^{\mathrm{O}}(t) - V^{\mathrm{S}}(t) > 0$ with zero capital at risk in expectation. Under risk neutrality, this is an arbitrage opportunity, and agents would buy sealed product until $V^{\mathrm{S}}(t)$ is bid up to $V^{\mathrm{O}}(t)$.

Conversely, suppose $\Delta(t) > 0$. An agent who can "construct" a sealed product (or equivalently, who is indifferent between the sealed product and the expected portfolio

of singles) would sell the sealed product and buy the equivalent portfolio of singles at expected cost $V^{\mathrm{O}}(t) < V^{\mathrm{S}}(t)$. This pushes $V^{\mathrm{S}}(t)$ down.

In equilibrium, $\Delta(t) = 0$. □

Remark 4.1. *This result mirrors standard financial theory. A sealed booster pack is economically equivalent to a lottery ticket over a portfolio of assets (individual cards). Under complete markets and risk neutrality, the price of the lottery equals the expected payoff. This is the CCG analog of risk-neutral pricing in derivatives markets: the sealed product is a derivative whose underlying assets are individual cards.*

Remark 4.2. *The argument for $\Delta(t) > 0$ is weaker than the argument for $\Delta(t) < 0$ because constructing a sealed product from singles is generally not possible—you cannot "re-seal" a booster. This asymmetry means that upward deviations ($\Delta > 0$, sealed premium) are harder to arbitrage away than downward deviations ($\Delta < 0$, sealed discount). We discuss this further in Section 5.*

5 Market Frictions and the EV Gap

In practice, $\Delta(t) \neq 0$. This section catalogs the primary frictions.

5.1 Transaction Costs

Selling individual cards on the secondary market involves costs:

- Platform fees (e.g., TCGPlayer charges sellers approximately 10–13% of the sale price).
- Shipping costs per order.
- Time and labor: listing, packaging, and mailing cards.

Let $\tau \in [0, 1)$ denote the effective proportional transaction cost. The *net expected value* of opening becomes

$$V^{\mathrm{O}}_{\mathrm{net}}(t) = (1 - \tau) \cdot V^{\mathrm{O}}(t). \tag{10}$$

Under this friction, the no-arbitrage condition becomes $V^{\mathrm{S}}(t) = (1 - \tau)\, V^{\mathrm{O}}(t)$, or equivalently,

$$\frac{\Delta(t)}{V^{\mathrm{O}}(t)} = \tau. \tag{11}$$

So we should expect the sealed product to trade at a premium equal to the transaction cost rate. If the typical fee is $\tau \approx 0.12$, then sealed product should trade at roughly 12% above the expected value of its contents—and indeed, this is a common empirical observation for in-print MTG products.

5.2 The Minimum-Value Threshold

Not all cards are worth selling. If a card's market price is below some threshold $\underline{p}$ (say, \$0.25), the transaction cost of selling it exceeds its value, so rational agents discard it. Define the *effective expected value* as

$$V_{\text{eff}}^{\text{O}}(t) = \sum_{k=1}^{n} \sum_{i=1}^{Z} \pi_{k,i} \cdot p_i(t) \cdot \mathbf{1}\{p_i(t) \geq \underline{p}\}. \tag{12}$$

Since most commons and many uncommons fall below $\underline{p}$, the effective expected value is strictly less than the theoretical expected value. This further widens the sealed premium.

5.3 Liquidity

Not all cards are equally liquid. A heavily played tournament staple has deep bid–ask markets and can be sold instantly, while a bulk rare has thin markets and may sit unsold for weeks. The *liquidity discount* reduces the effective realizable value of low-demand cards.

Let $\lambda_i(t) \in [0, 1]$ be the liquidity factor for card c_i at time t, where $\lambda_i = 1$ means perfectly liquid and $\lambda_i = 0$ means unsellable. The liquidity-adjusted expected value is

$$V_{\lambda}^{\text{O}}(t) = \sum_{k=1}^{n} \sum_{i=1}^{Z} \pi_{k,i} \cdot \lambda_i(t) \cdot p_i(t). \tag{13}$$

5.4 Variance and Risk Aversion

Even under zero transaction costs, a risk-averse agent values the lottery at less than its expected value. The value of opening a single product has variance

$$\text{Var}\left[\sum_{k=1}^{n} X_k(t)\right] = \sum_{k=1}^{n} \text{Var}[X_k(t)] + 2\sum_{k<j} \text{Cov}[X_k(t), X_j(t)]. \tag{14}$$

For a typical booster, the variance is dominated by the rare/mythic slot. If a set has a single "chase mythic" worth \$50 while most rares are worth \$1, the variance of the rare/mythic slot is very high. A risk-averse buyer would demand a discount on the sealed product to compensate for this variance, which would push $\Delta(t)$ negative.

However, MTG sealed products also have a *gambling premium*: many buyers enjoy the variance itself (the thrill of opening a valuable card). This behavioral premium can push $\Delta(t)$ positive, analogous to the observation that lottery tickets systematically trade above expected value.

5.5 Irreversibility and the Sealed Premium

As noted in Section 4, opening a sealed product is irreversible. You cannot reconstruct a factory-sealed booster from individual cards. This creates an asymmetry:

- If $\Delta(t) < 0$ (sealed discount), agents can arbitrage by buying sealed, opening, and selling singles.
- If $\Delta(t) > 0$ (sealed premium), arbitrage requires selling sealed and buying singles—but you cannot deliver a sealed product assembled from singles.

The irreversibility of opening means sealed products can sustain a premium above expected value. The sealed product has an *option value*: the holder retains the right to open it later, when card prices may have changed. Keeping the product sealed preserves optionality, while opening eliminates it. This is analogous to the time value of an American option: early exercise (opening) sacrifices future optionality.

5.6 Summary of Frictions

Combining all frictions, the equilibrium spread satisfies

$$\Delta(t) = \underbrace{\tau \cdot V^{\mathrm{O}}(t)}_{\text{transaction costs}} + \underbrace{\left(V^{\mathrm{O}}(t) - V^{\mathrm{O}}_{\mathrm{eff}}(t)\right)}_{\text{unsellable cards}} + \underbrace{\text{liquidity discount}}_{\text{thin markets}} + \underbrace{\text{option value}}_{\text{irreversibility}} + \underbrace{\text{gambling premium}}_{\text{behavioral}} - \underbrace{\text{risk}}_{\text{varian}} \tag{15}$$

The sign of $\Delta(t)$ depends on the relative magnitude of these forces. For in-print products with low secondary market value, the transaction cost and unsellable-card effects dominate, and $\Delta(t) > 0$ (sealed premium). For out-of-print products with high-value singles, the spread can turn negative ($\Delta < 0$), making it profitable to buy sealed and open.

6 Time Dynamics

Both $V^{\mathrm{S}}(t)$ and $V^{\mathrm{O}}(t)$ evolve over time. Their dynamics are linked but not identical.

6.1 Card Price Dynamics

Individual card prices $p_i(t)$ respond to:

- **Tournament results**: a card that wins a major tournament spikes in price.
- **Ban-list changes**: a banned card crashes; cards that benefit from the ban rise.
- **Set rotation**: cards leaving Standard format typically decline; cards entering Modern/Legacy may rise.
- **Supply changes**: reprints in new products increase supply and depress price.
- **Speculative activity**: buyouts, hoarding, and panic selling.

Without imposing a specific stochastic process on $p_i(t)$, we note that the expected value of the product inherits the price dynamics of its contents:

$$\frac{dV^{\mathrm{O}}(t)}{dt} = \sum_{k=1}^{n} \sum_{i=1}^{Z} \pi_{k,i} \cdot \frac{dp_i(t)}{dt}. \tag{16}$$

The rate of change of the opened value is a probability-weighted average of the rates of change of individual card prices. If the "chase cards" appreciate while bulk depreciates, the expected value may rise or fall depending on which effect dominates.

6.2 Sealed Price Dynamics

The sealed price $V^{\mathrm{S}}(t)$ responds to the same forces as individual card prices, plus:

- **Supply**: sealed supply decreases over time as product is opened or consumed. Once a set goes out of print, sealed supply is fixed and declining.
- **Collector demand**: sealed product has value to collectors independent of its contents. Nostalgia, completionism, and display value all contribute.
- **Reprint risk**: the possibility that the same cards will be reprinted in a future product reduces the option value of the sealed product.

For long-dated out-of-print products, sealed prices often diverge dramatically from expected opened value. A sealed booster box from 1994–1996 era Magic may trade at thousands of dollars, far exceeding the expected value of the cards inside, because the sealed product itself has become a collectible.

6.3 The Spread as a Time Series

Define the *relative spread*:

$$\delta(t) = \frac{\Delta(t)}{V^{\mathrm{O}}(t)} = \frac{V^{\mathrm{S}}(t)}{V^{\mathrm{O}}(t)} - 1. \tag{17}$$

For in-print products, $\delta(t)$ is typically bounded and mean-reverting: if card prices spike (increasing V^{O}), sealed prices adjust upward with a lag, temporarily compressing δ. Conversely, if the set is "opened out" and singles prices crash, sealed prices may hold up better, widening δ.

For out-of-print products, $\delta(t)$ tends to increase monotonically over time as the sealed premium accrues collector value.

7 Generalization Across Product Types

The framework accommodates various MTG product types by varying n and $\{\boldsymbol{\pi}_k\}$.

7.1 Draft Booster

$n \approx 15$. Slot distributions follow rarity rules. High variance, concentrated in the rare/mythic slot. Expected value typically below sealed price for in-print product.

7.2 Collector Booster

$n \approx 15$, but slots contain only rares, mythics, foils, and special treatments. The probability distributions $\boldsymbol{\pi}_k$ assign zero weight to commons and uncommons. Higher expected value per slot, but also higher sealed price. Variance is typically higher because the entire pack consists of high-variance slots.

7.3 Preconstructed Deck (Commander)

$n = 100$ (or $99 + 1$ commander). Contents are deterministic: $\pi_{k,i} \in \{0, 1\}$. There is no randomness, so

$$V^{\mathrm{O}}(t) = \sum_{k=1}^{100} p_{i(k)}(t),$$

where $i(k)$ is the specific card in slot k. The spread $\Delta(t)$ in this case reflects pure transaction costs, liquidity, and the convenience of buying a pre-assembled deck versus acquiring 100 singles.

7.4 Booster Box

A booster box contains B boosters (typically $B = 36$ for draft or $B = 12$ for collector boosters), and the expected value scales linearly:

$$V^{\mathrm{O}}_{\mathrm{box}}(t) = B \cdot V^{\mathrm{O}}_{\mathrm{booster}}(t). \tag{18}$$

However, collation within a box introduces correlation: boxes are typically collated so that each rare appears at most once or twice. This reduces variance relative to B independent boosters but does not change the expected value (again by linearity of expectation).

The box-level spread

$$\Delta_{\mathrm{box}}(t) = V^{\mathrm{S}}_{\mathrm{box}}(t) - B \cdot V^{\mathrm{O}}_{\mathrm{booster}}(t) \tag{19}$$

can differ from $B \cdot \Delta_{\mathrm{booster}}(t)$ if box-level pricing reflects a volume discount or if the collation guarantee (e.g., "at least one mythic per box") alters expected value.

8 A Generic Numerical Example

Consider a hypothetical set $\mathcal{S}$ with the following structure:

Table 1: Set composition and average prices.

Rarity	**Count**	**Avg. Price**	**Max Price**	**Min Price**
Common ($\mathcal{C}$)	101	\$0.05	\$0.25	\$0.02
Uncommon ($\mathcal{U}$)	80	\$0.15	\$2.00	\$0.03
Rare ($\mathcal{R}$)	60	\$1.50	\$12.00	\$0.25
Mythic Rare ($\mathcal{M}$)	20	\$6.00	\$40.00	\$0.50

Draft booster ($n = 15$: 1 rare/mythic, 3 uncommon, 10 common, 1 bonus slot). Assume the bonus slot contributes an expected \$0.20.

Using (8):

$$\begin{aligned} V^{\mathrm{O}} &= \frac{7}{8}(1.50) + \frac{1}{8}(6.00) + 3(0.15) + 10(0.05) + 0.20 \\ &= 1.3125 + 0.75 + 0.45 + 0.50 + 0.20 \\ &= \$3.21. \end{aligned}$$

If the sealed booster retails at $V^{\mathrm{S}} = \$4.49$, the spread is

$$\Delta = 4.49 - 3.21 = \$1.28, \qquad \delta = \frac{1.28}{3.21} \approx 39.9\%.$$

This spread is explained by transaction costs ($\sim$12%), unsellable bulk ($\sim$10%), and the gambling/sealed premium ($\sim$18%). The retailer's margin is embedded in the sealed price, which is a real-world transaction cost absent from the theoretical frictionless benchmark.

Now suppose the set contains one "chase mythic" worth \$40. The expected contribution of that single card to the rare/mythic slot is

$$\frac{1}{8} \cdot \frac{1}{20} \cdot 40 = \$0.25.$$

That single card contributes about 7.8% of the entire booster's expected value. If that card's price doubles to \$80 (due to a tournament win), the booster's expected value increases by \$0.25 to \$3.46, and the sealed price would adjust upward with some lag.

8.1 Variance Illustration

The variance of the rare/mythic slot is

$$\mathrm{Var}[X_{\text{rare slot}}] = \mathbb{E}[X^2_{\text{rare slot}}] - (\mathbb{E}[X_{\text{rare slot}}])^2.$$

If rare prices are distributed with mean \$1.50 and second moment $\mathbb{E}[p_R^2] = \frac{1}{60}\sum_{c_i \in \mathcal{R}} p_i^2$, and similarly for mythics, the variance can be substantial. For the numbers in Table 1, a rough calculation gives $\mathrm{Var}[X_{\text{rare slot}}] \approx$ \$7–\$15 (in squared dollars), implying a standard deviation of \$2.50–\$3.90 for a product with expected value \$3.21. The coefficient of variation exceeds 1, confirming that booster opening is a high-variance activity.

9 Discussion

9.1 Efficient Markets in the MTG Secondary Market

The no-arbitrage condition $\Delta(t) = 0$ is the MTG analog of the efficient market hypothesis. In an efficient CCG secondary market, sealed product prices fully reflect the expected value of their contents, adjusted for transaction costs. Deviations represent either genuine market frictions or mispricings that create profit opportunities for informed participants.

Empirically, the MTG secondary market is semi-efficient at best. Large, liquid cards on major platforms are priced competitively, but small-market cards, foreign-language printings, and niche foils exhibit persistent mispricings. The sealed market is even less efficient: retail prices are sticky, distributor pricing involves multi-tier markups, and collector demand introduces pricing factors orthogonal to card-level fundamentals.

9.2 The Frequentist Probability Assumption

Throughout this paper, we treat the slot-level probabilities $\pi_{k,i}$ as known. In practice, these are estimated from the product's design rules (which Wizards of the Coast publishes or which can be reverse-engineered from large opening samples). The frequentist assumption—that each card of a given rarity is equally likely—is approximately correct for most MTG products but breaks down for:

- Products with curated card lists (e.g., Secret Lair drops with fixed contents).
- Print sheets with non-uniform layouts (some mythics appear on print sheets at different rates).
- Products with bonus sheets or special slots that follow non-uniform distributions.

When the exact distribution is unknown, the expected value $V^{\mathrm{O}}(t)$ becomes a point estimate subject to model uncertainty. Bayesian approaches could be used to quantify this uncertainty, but we leave that extension for future work.

9.3 Implications for Market Participants

The model has practical implications:

- **Retailers**: sealed product pricing should track $V^{\mathrm{O}}(t)$ plus a markup reflecting transaction costs and convenience. When singles prices crash, retailers face margin

compression on sealed inventory.

- **Speculators**: monitoring the relative spread $\delta(t)$ across products identifies opportunities. A product with abnormally low δ may be underpriced sealed; one with abnormally high δ may be overpriced.
- **Consumers**: the expected loss from opening a booster at retail is approximately $\Delta(t)$. This is the "entertainment premium" paid for the experience of opening.

10 Conclusion

We have presented a valuation framework for sealed CCG products that links the sealed product price to the expected value of its randomly determined contents. The core result is a no-arbitrage condition: under perfect markets, the sealed price equals the expected opened value. In practice, transaction costs, liquidity frictions, variance, irreversibility, and behavioral premia drive a persistent wedge—the sealed–opened spread.

The framework is intentionally kept at the level of applied probability and basic financial theory. Extensions to stochastic price processes, dynamic programming formulations of optimal opening timing, and empirical estimation of the spread using real market data are natural next steps.

For the practitioner, the key takeaway is simple: *the sealed premium is the price of optionality, convenience, and entertainment.* In an imperfect market, this premium is positive for in-print product, reflecting the real costs of converting a sealed product into cash through the singles market. For out-of-print product, collector demand and declining supply can push the premium far beyond what frictions alone would justify.

References

[1] Hull, J.C. (2018). *Options, Futures, and Other Derivatives.* 10th ed. Pearson.

[2] Varian, H.R. (2014). *Intermediate Microeconomics: A Modern Approach.* 9th ed. W.W. Norton.

[3] Fama, E.F. (1970). Efficient capital markets: A review of theory and empirical work. *Journal of Finance*, 25(2), 383–417.

[4] Thaler, R.H. and Ziemba, W.T. (1988). Anomalies: Parimutuel betting markets—Racetracks and lotteries. *Journal of Economic Perspectives*, 2(2), 161–174.

[5] Ross, S.A. (1976). The arbitrage theory of capital asset pricing. *Journal of Economic Theory*, 13(3), 341–360.